



Classifications of DECam Observed Transients using ParSNIP

Jack Collard, Yaamini Jois, Nathan Olson, Orestis Sabetai, Xavier Silva, Mathew Thomas, Mentor: Emilie Cote

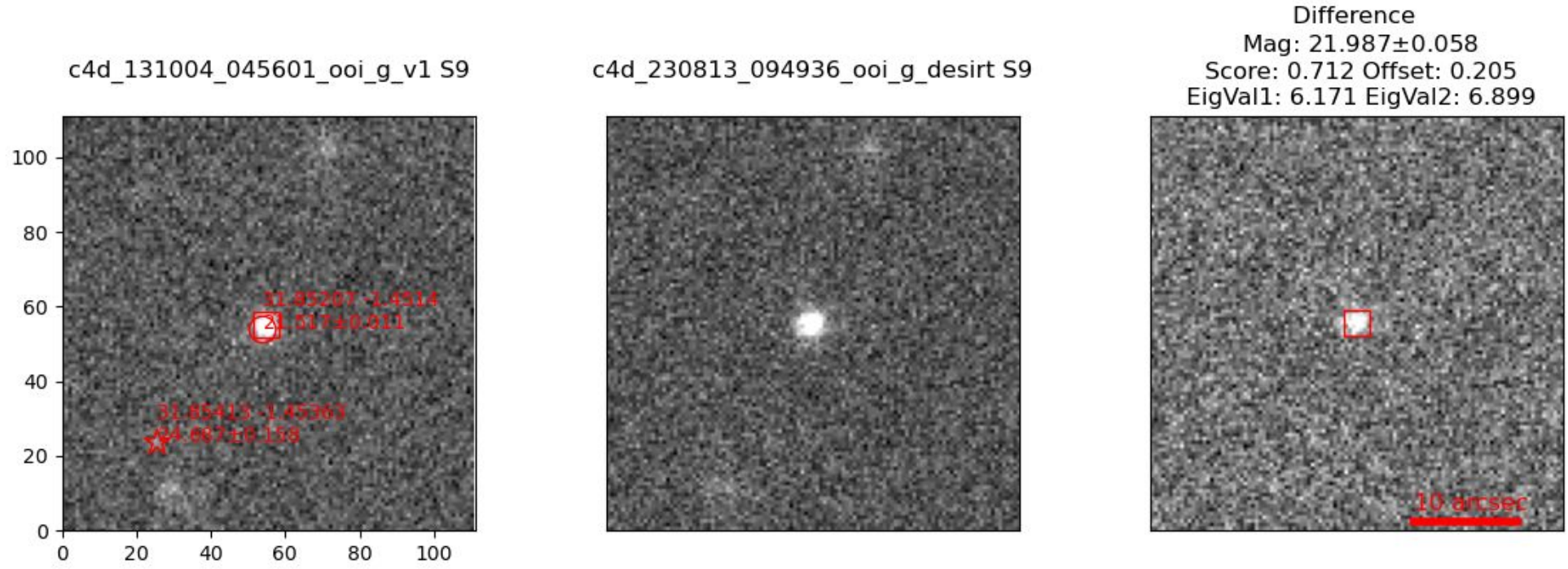
University of California, Berkeley Undergraduate Lab at Berkeley, Physics & Astronomy Division



BACKGROUND

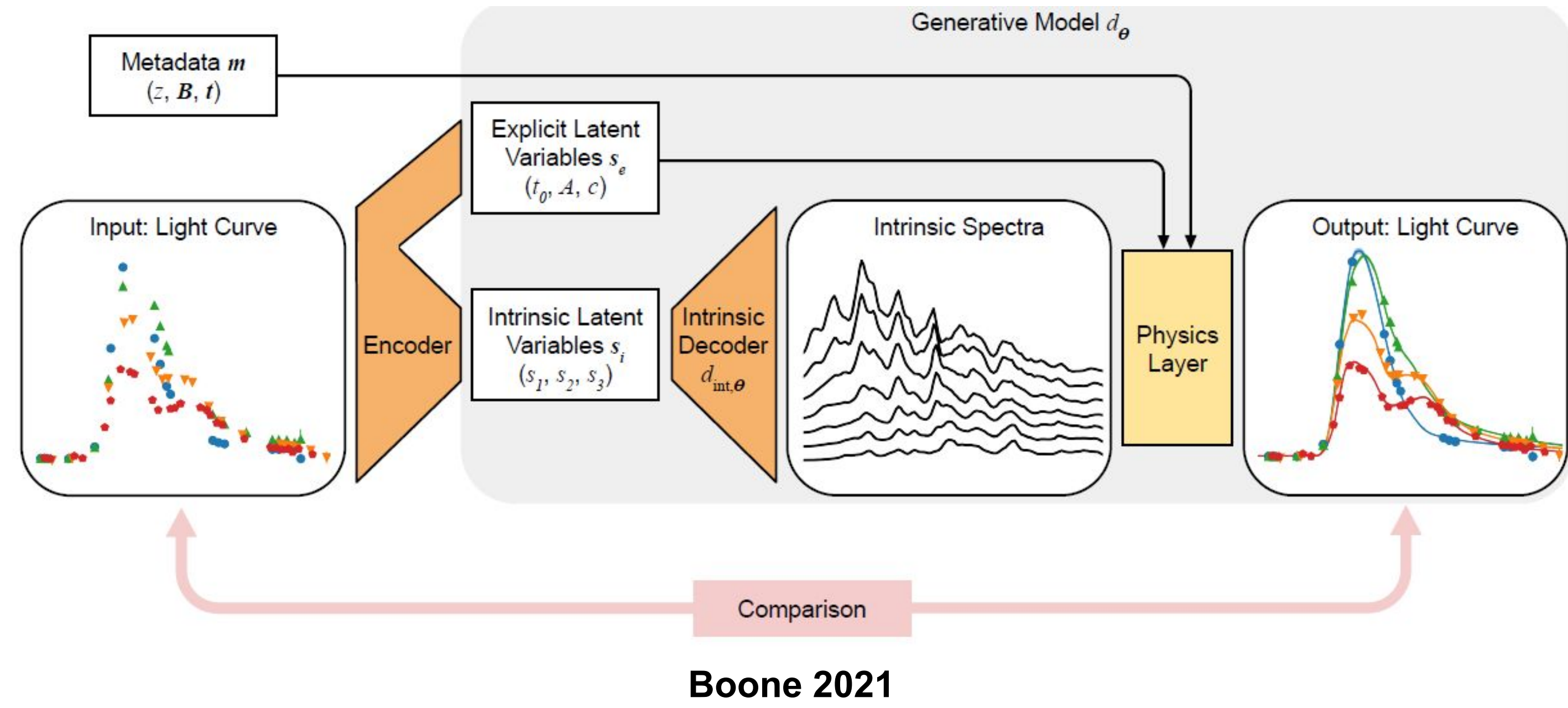
The term “**transients**” encompasses a variety of galactic and extragalactic events, as they are astronomical phenomena with durations ranging from a fraction of a second to years. The classification of transients requires sorting through lots of complex and unlabeled data, leaving room for human error. Traditionally, classification is done through tedious spectroscopic observations, but this isn't feasible with how many transients we can observe now. Dimmer transients are also much harder to classify by hand. **ParSNIP (Parametrization of Supernova Intrinsic Properties)**, is a generative neural network that generates models that simplify the classification of transients and increase classification accuracy without the need for labeled data. The received data can change in value depending on how it was taken even for identical transient types, and eliminating the influence of external factors can be done efficiently with ParSNIP. In this way, dimmer transients can also be classified without traditional spectroscopy. ParSNIP trains itself using the simulated datasets we feed it to improve its accuracy for predictions.

One of over 50 observations for Example 1 light curve, SNII object



METHODS – PARSNIP

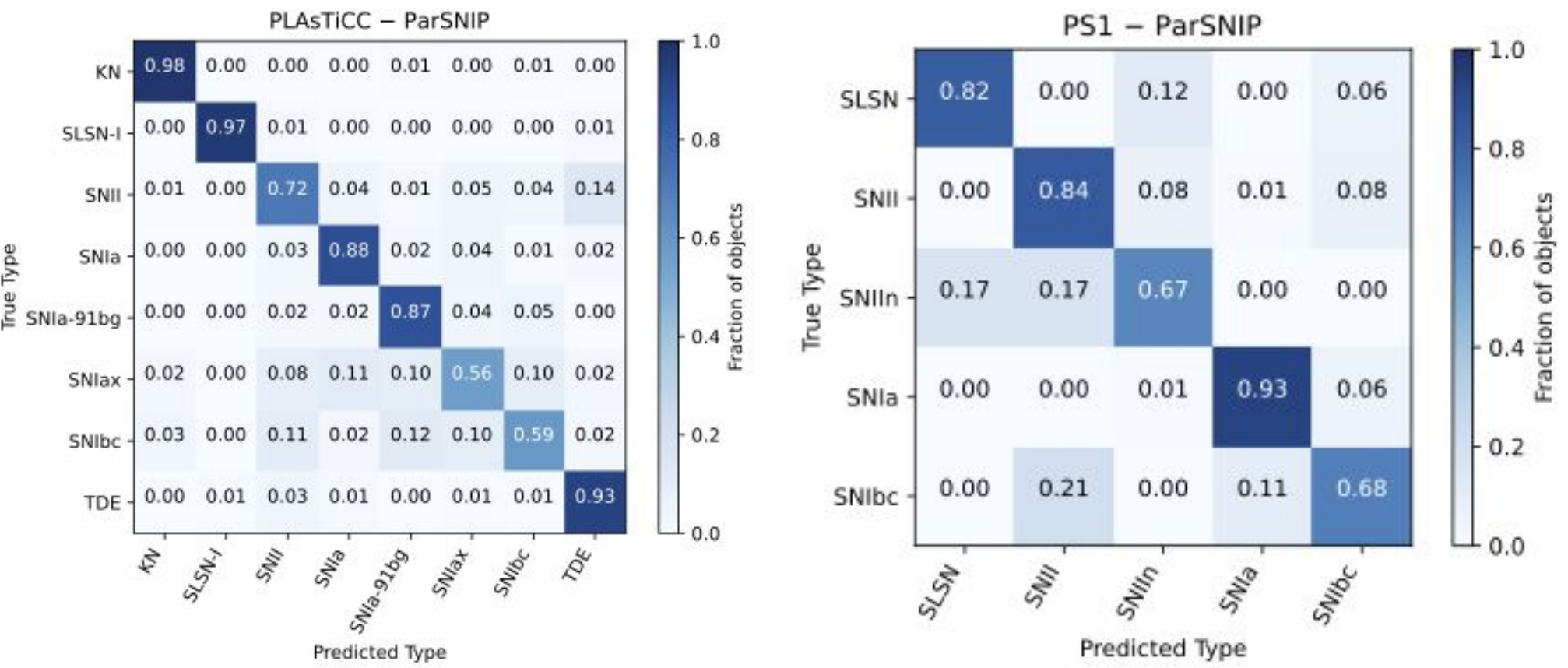
ParSNIP uses a variational autoencoder, so it takes the data we input, compresses it, and places it in a latent vector space composed of latent variables. ParSNIP produces generative models of transients from datasets of unlabeled light curves. We obtained data from the **DECam joint** with LSST and DESI in order to make predictions and classifications with ParSNIP. Raw data had to be molded into **lcdata** format for project use, forming meta tables with object IDs and uniform observation **bands** (r, g, z).



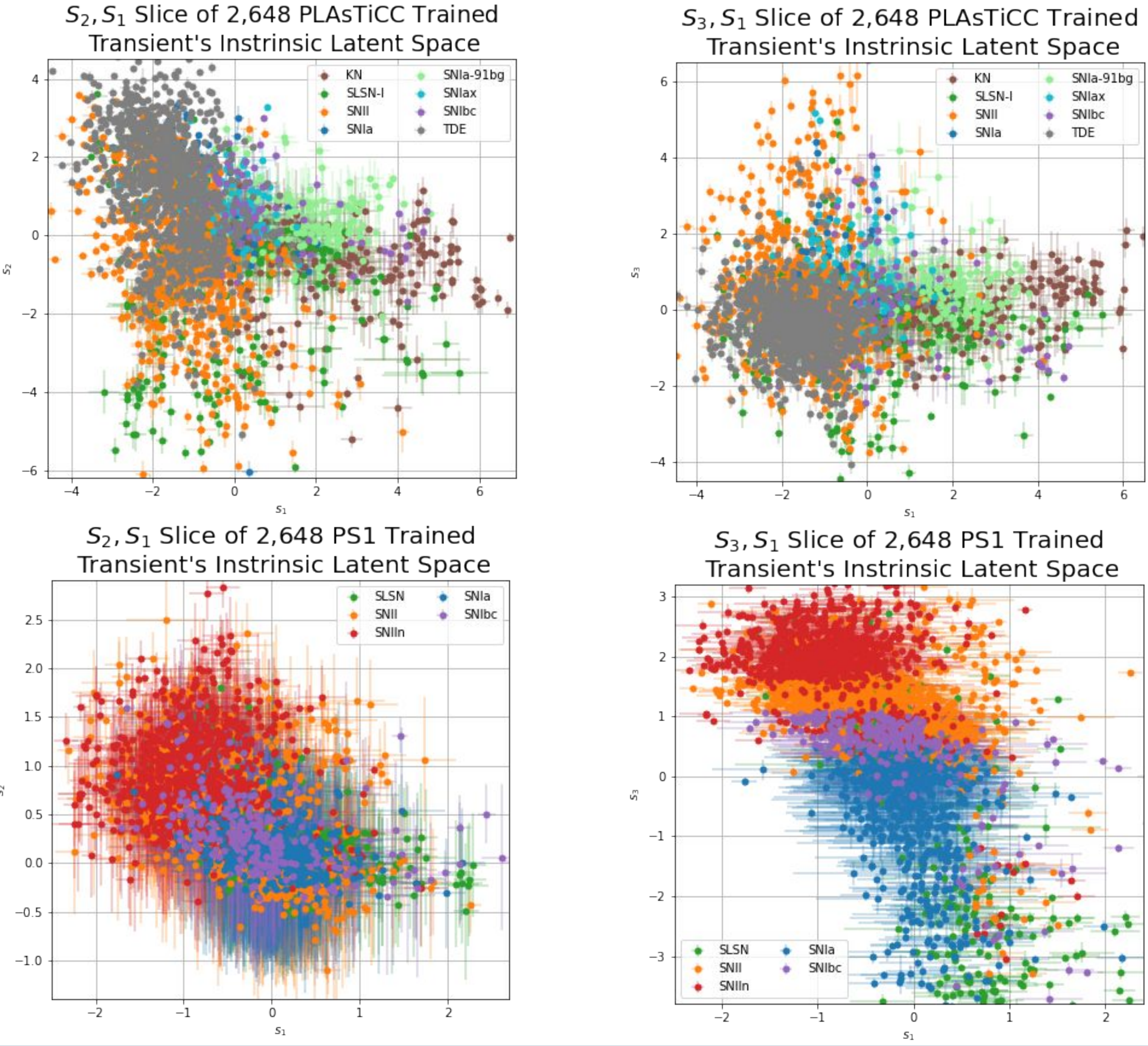
Boone 2021

ParSNIP was trained on labeled data from two sources: first, from simulated light curve data done by **PLAsTiCC (Photometric LSST Astronomical Time-Series Classification Challenge)** to mimic data from the LSST at Vera C. Rubin Observatory, and second, from real data taken by **Pan-STARRS1 (PS1)**.

Ideal confusion matrices for both datasets



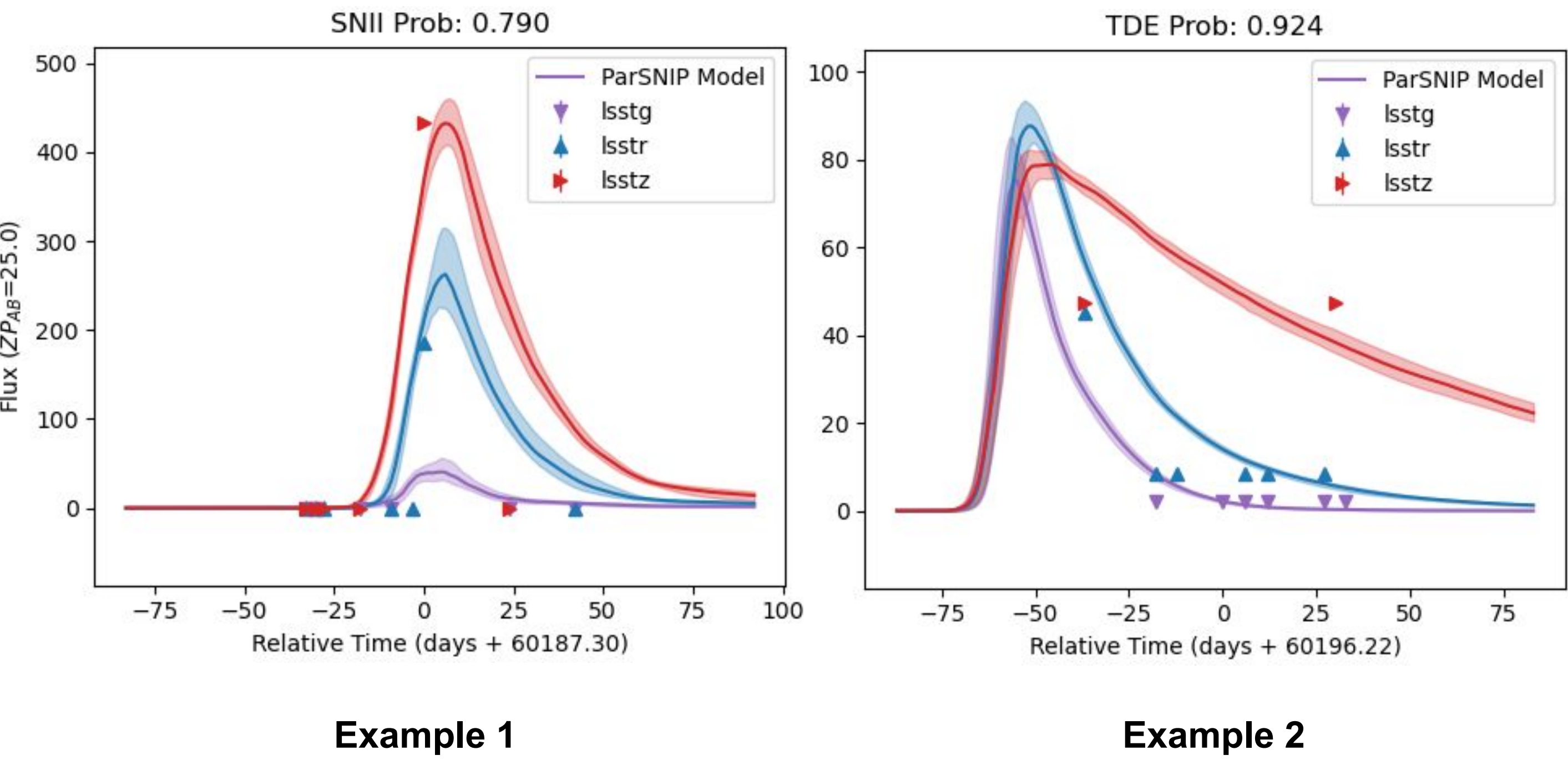
DATA



RESULTS

The results we obtained are an outcome of using a built-in ParSNIP function that takes a data file and a model and uses the data points to plot a lightcurve that best fits the data it was given. Below, Example 1 shows an example of an acceptable **lightcurve**, where the model fits the data points well, and the error (shown by the faded color) is minimal and fits closely to the model line. Example 2 shows a lightcurve that displays a limitation in the data we have: often, the **flux** stays consistent over time, raising questions about whether the observed “transient” is a transient at all.

$$\text{Flux} = 10^{\frac{22.5 - \text{Magnitude}}{2.5}} \quad \text{Where, 22.5 is the Zero Point (Z.P.)}$$



RESULTS

We used five datasets each with a varying number of transients and redshifts to produce multiple stats tables. The following stats tables contained the least possible error values and came from a dataset that contained 2267 transients of both photometric and spectroscopic **redshifts** with at least 15 observations as a filter (2648 total.) Each classification is associated with a certain possibility of accuracy, along with the number of objects that are predicted to have at least an 85% possibility of being a certain transient, with the lower threshold of possibility being 50% for PS1 data and 55% for PLAsTiCC data. This analysis is also how we can catch potential errors, as certain transients should not, theoretically, be as present as ParSNIP may predict it to be.

Type	Total	p > 0.85	Highest Prob	Lowest Prob	Mean Prob	Median Prob	Max Data pts	Min Data pts	Ave Data pts
str5	int64	int64	float64	float64	float64	float64	int64	int64	int64
SLSN	199	67	0.98	0.504	0.772	0.805	94	16	45
SNI	479	54	0.99	0.5	0.645	0.599	96	16	46
SNIIn	648	73	0.992	0.501	0.677	0.655	96	16	49
SNIa	364	112	0.978	0.503	0.762	0.779	103	16	48
SNIbc	86	4	0.932	0.5	0.625	0.608	96	17	46

PS1 Results

Type	Total	p > 0.85	Highest Prob	Lowest Prob	Mean Prob	Median Prob	Max Data pts	Min Data pts	Ave Data pts
str6	int64	int64	float64	float64	float64	float64	int64	int64	int64
KN	172	99	0.999	0.555	0.853	0.88	97	17	50
SLSN-I	245	127	0.998	0.55	0.83	0.863	97	16	46
SNI	544	247	0.997	0.551	0.809	0.815	99	16	48
SNIa	22	3	0.957	0.571	0.732	0.731	84	16	45
SNIax	35	6	0.926	0.563	0.724	0.723	96	17	42
SNIbc	46	10	0.99	0.55	0.728	0.716	94	18	38
TDE	729	503	0.997	0.551	0.881	0.944	103	16	46

PLAsTiCC Results

CONCLUSION

Over an academic year, we accomplished the broader goals of successfully running ParSNIP to produce light curves, stats tables, and latent spaces as an extension and hopeful improvement to the older version of the project. However, we were unable to achieve the high accuracy we aimed for within our predictions. This occurred as a result of our group and resources not being able to distinguish why our data showed magnitudes that did not match up with the predicted light curves that ParSNIP generated. There is also the possibility that ParSNIP itself was not properly trained on the datasets that we were drawing from (Dark Energy Camera (DECam) Legacy Survey). Due to our lack of time to run enough code to properly train our model and figure out the individual bands within each dataset, we were not able to properly identify why the data points were plotted so inconsistently. Since there is room for future improvement and continuation of this project, we received access to another dataset in the final week of our presentation as well as proposals for side projects to utilize the dataset, which has also been visualized in the updated slide and stats tables. In future work, this dataset will provide more paths for the improvement of the network.

ACKNOWLEDGEMENTS

We would like to thank ULAB Physics and Astronomy for supporting our project: Directors **Rav Kaur** and **Saahit Mogan**, Lab Manager **Jordan Duan**, and Faculty Advisor **Professor Dan Kasen**. We would also like to thank the Legacy Survey Project and the collaboration between the Dark Energy Camera and DESI for sourcing all the datasets we used throughout this project.

REFERENCES

- Kyle Boone (2021). “ParSNIP: Generative Models of Transient Light Curves with Physics-enabled Deep Learning”
- Young Supernova Experiment(2022). “Experiment Data Release 1 (YSE DR1): Light Curves and Photometric Classification of 1975 Supernovae